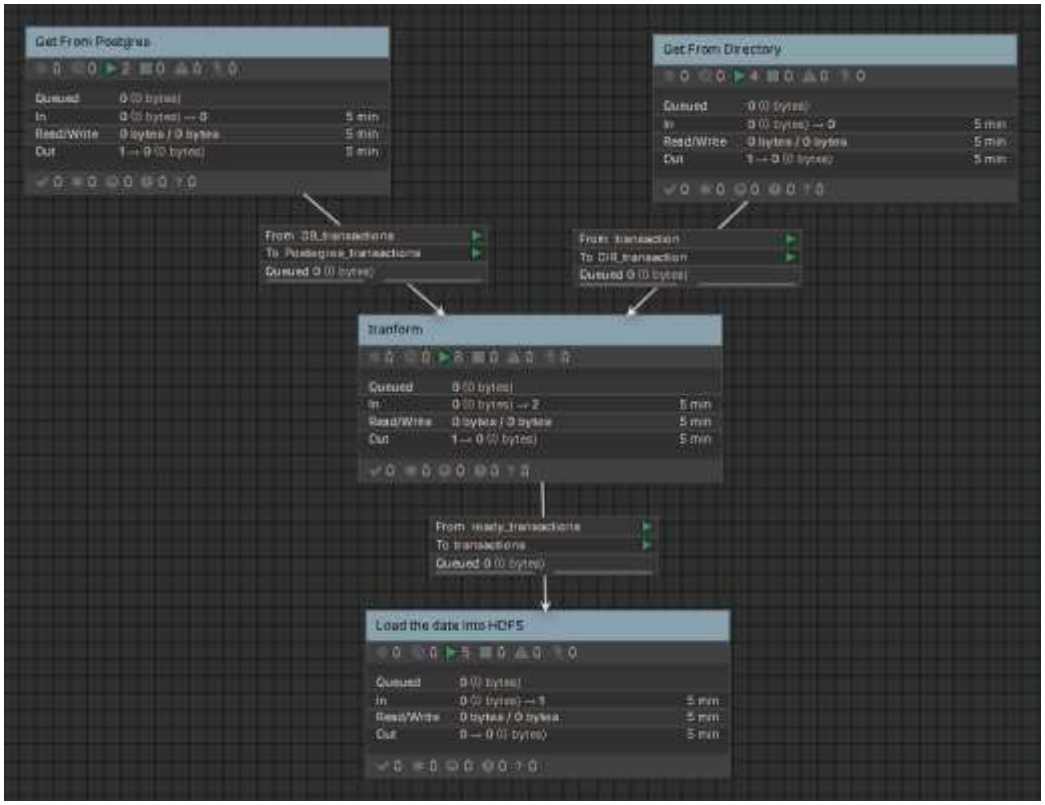
Odai Aqlan

Architecture Diagram



I divide the project into those four process groups and connect them with input and output ports to keep the flow organized and scalable.

Get from Postgres group to pull data from the database.

Get from Directory group to read files from a folder.

Transformer group to clean and prepare the data.

Load the data into HDFS group to store the processed data in Hadoop Distributed File System.

Ask me Why?

Because structuring the flow this way, I make it easier to maintain, extend, and scale later and it’s common best practise .

I will explain each process group and the decisions down there ,,,

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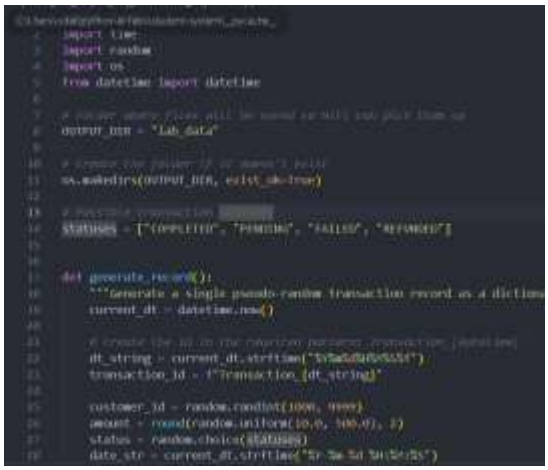
Real-Time Data Simulation

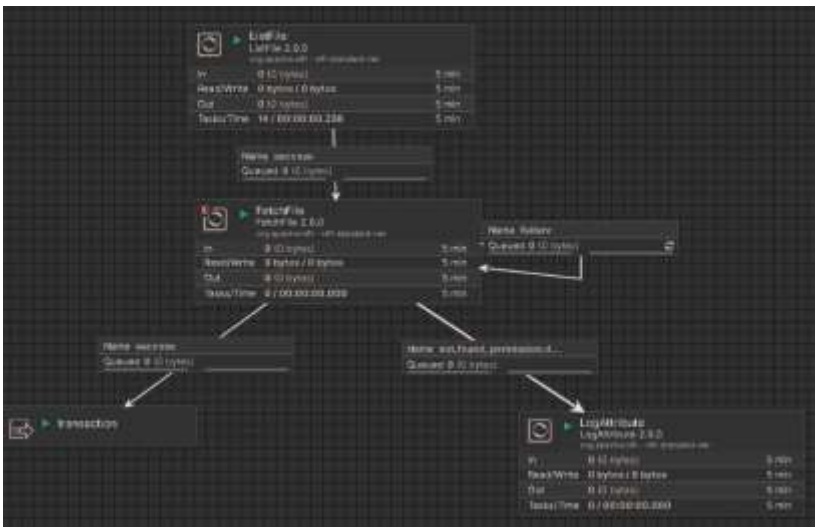
I created python script to genrate dublicats messy json files every 5 sec that simulate real world Transactions and also save it in directory



The secript in the same folder out there

File-Based Ingestion using NiFi

This is the first process group Get from Directory



List file processor : I used it as observer when there are any new Transactions till the next process to fetch it and it runs every 5 sec .

FetchFile processor :I used it as hunter when the observer tell him there is file then he fetch it and it runs every time.

Also Log attribute to log not found and permission denied files ,,

but for falure i make it retry after penlized duarition ,

for success files its go to the next layer

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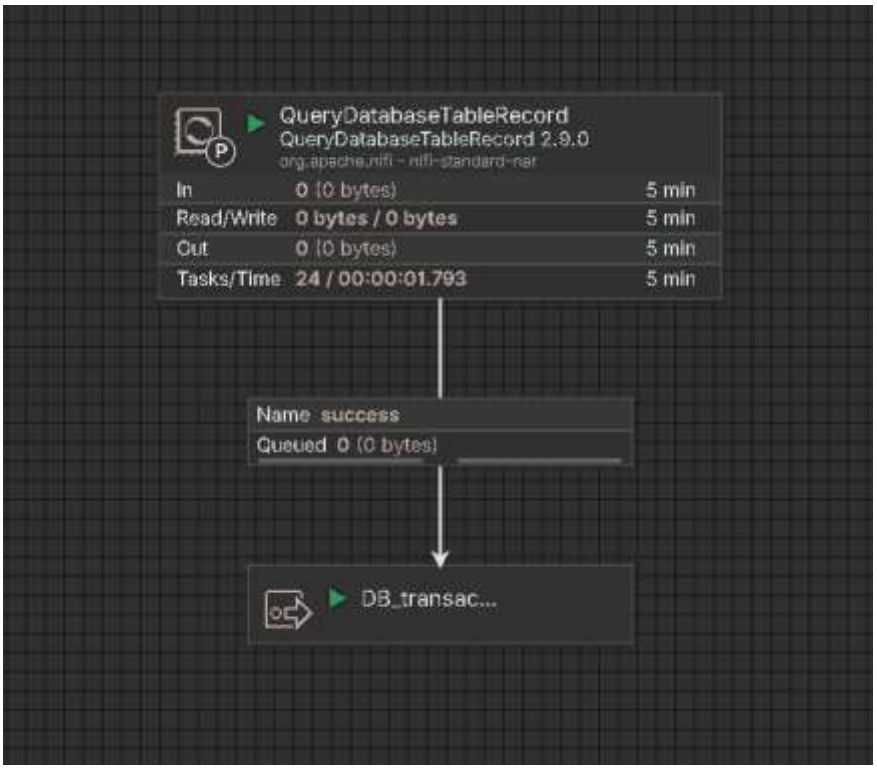
Database Ingestion

This is the second process group Get from Postgres



in this group I used the QueryDatabaseTableRecurd to to get the data from pstegres database every 5 sec because server re resources

ask me why QueryDatabaseTableRecurd but not QueryDatabaseTable or ExecuteSQL?

The main Big reason is :its directly converts database rows into json or any data type which that what I want the most beside That I can set format schema in JsonRecordSetWriter and because the coulmns names in DB not the same as the json files and to simplify the transformer layer which is the next

Other reason : I want it fetch only the new records which It can handle this part effectively .

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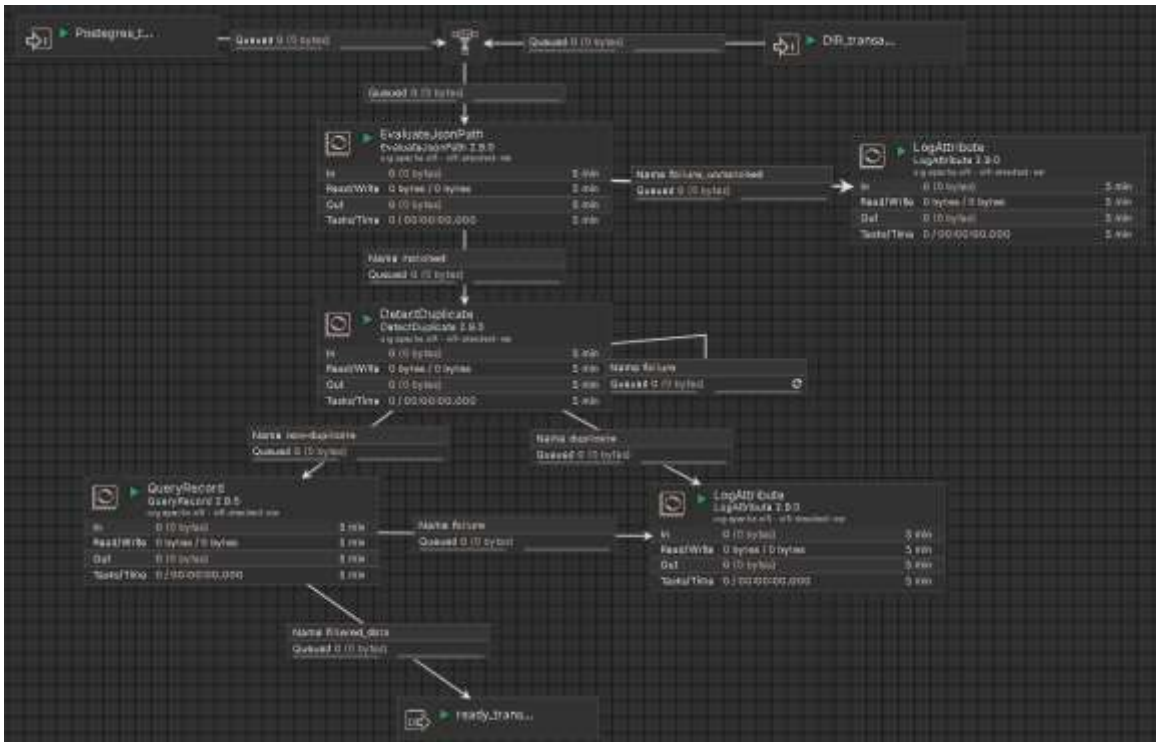
Data Transformations

This is the third process group Transformer



Because of the JsonRecordSetWriter and when I use it in the QueryDatabaseTableRecurd the data come from the two source with the same schema , so now I can use the same flow for all data whatever the source is ,

The data come from two input ports from directory and pstegres I set the Prioriti for the older records

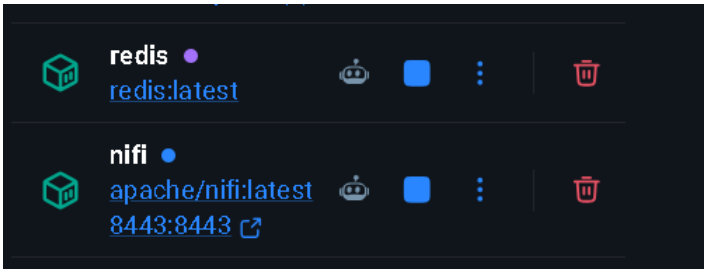
and I set the Back Pressure on 3000 flowfile to handle the peak hours

the Run Schedule :this layer or process group every time is runing because its important that not stop working

EvaluateJsonPath : I used it to read the flowfile and get the transaction_id as Aturbute because I need it in next process

DetectDuplicate : i used this process to detect the duplicated transactions because that cuase confilct in the data

But why I chose this process not others ?



I chose DetectDuplicate processor to prevents any record with the same ID from passing through more than once.the main feture is I link it to Redis as an external DistributedMapCache to sync across multiple NiFi nodes simultaneously in future which ensure integrity for my data

QuerRecord : simply I used it to ensure there is any vaule is null in record .

logAttribute : for log all failurs , dublicated and unmatched transactions

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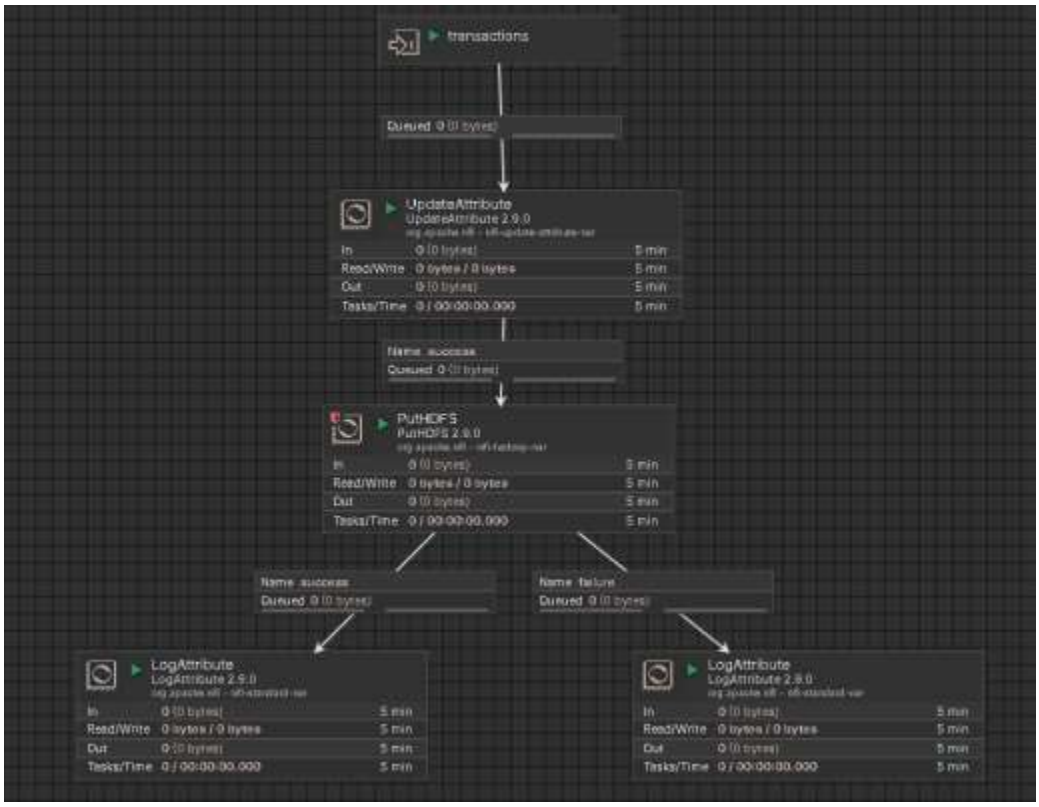
Load to HDFS

This is the third process group Load the data into HDFS



I get the transactions from the previous group process (transformer) as clean non-duplicates trans

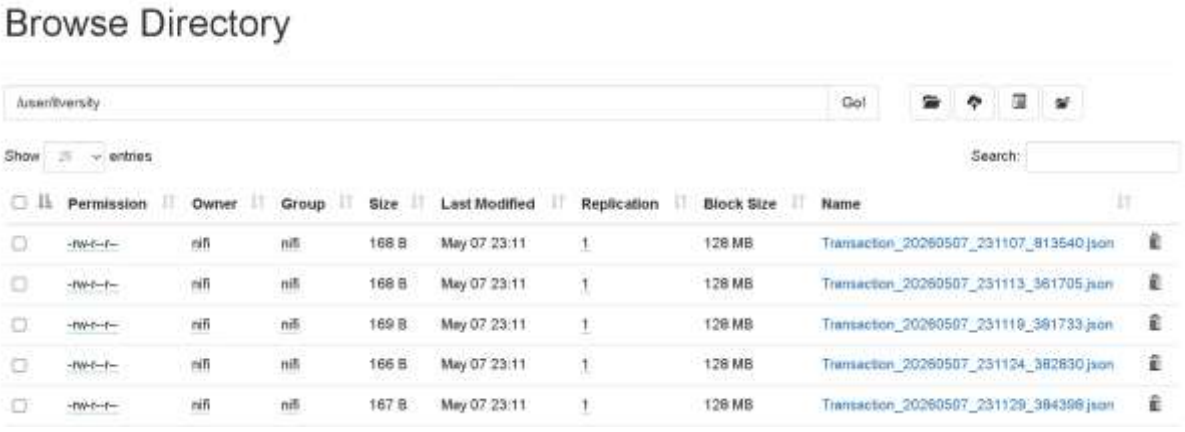
So now I have to store them in Hadoop file system

Update Attribute : I used it to creeate atturbute contain the file name acctualy the process genrate it

Put HDFS : to connect the hdfs server and save the transactions destrputed in file system

logAttribute : to log all the status

now tha data is ready



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Sample data before and after transformation

Before


```
[
  {
    "Transaction_ID": "Transaction_20260507202822747162",
    "Customer_ID": null,
    "Amount": null,
    "Status": "REFUNDED",
    "Date": "2026-05-07 20:28:22"
  },
  {
    "Transaction_ID": "Transaction_1001",
    "Customer_ID": 88,
    "Amount": 250.00,
    "Status": "SUCCESS",
    "Date": "2026-05-08 12:00:00"
  },
  {
    "Transaction_ID": "Transaction_1001",
    "Customer_ID": 88,
    "Amount": 250.00,
    "Status": "SUCCESS",
    "Date": "2026-05-08 12:00:00"
  }
]
```


after